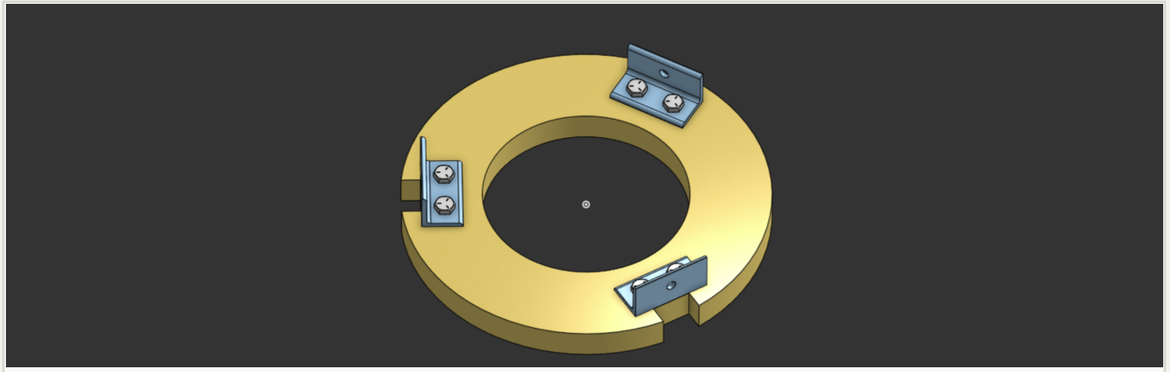
01 / SELECTED WORK

Engineering in practice.

Design decisions, physical hardware, and hands-on testing across team and personal projects.

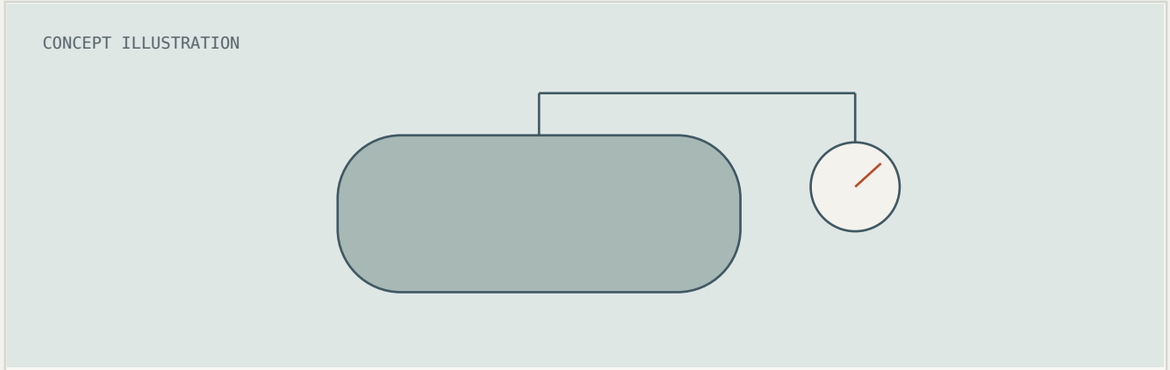


BEACH LAUNCH TEAM / 2026 - PRESENT

Bulkhead design & optimization

Structural trade studies and manufacturable CAD balancing load paths, mass, and routing space.

STRUCTURAL ANALYSIS / CAD / DFM

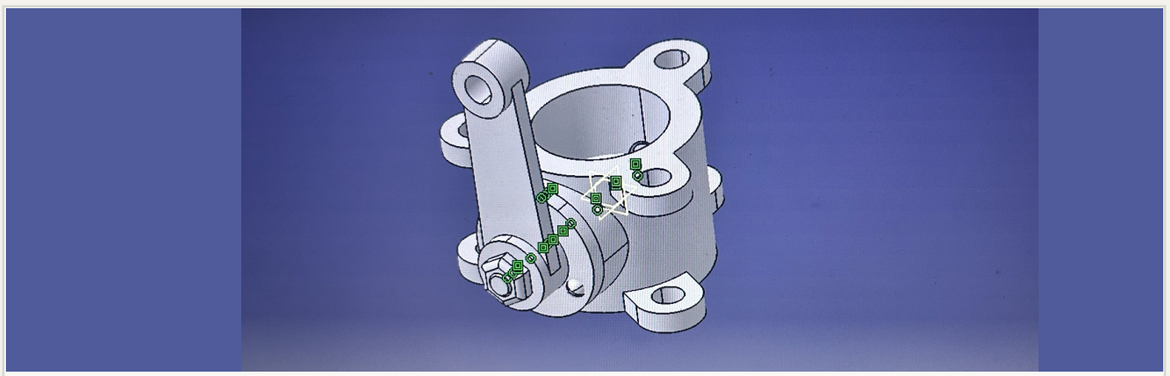


BEACH LAUNCH TEAM / 2026 - PRESENT

COPV hydrostatic proof testing

Operating high-pressure pumping systems and logging real-time data during a 4,500 psi proof test.

HYDROSTATIC TESTING / DAQ / COPV

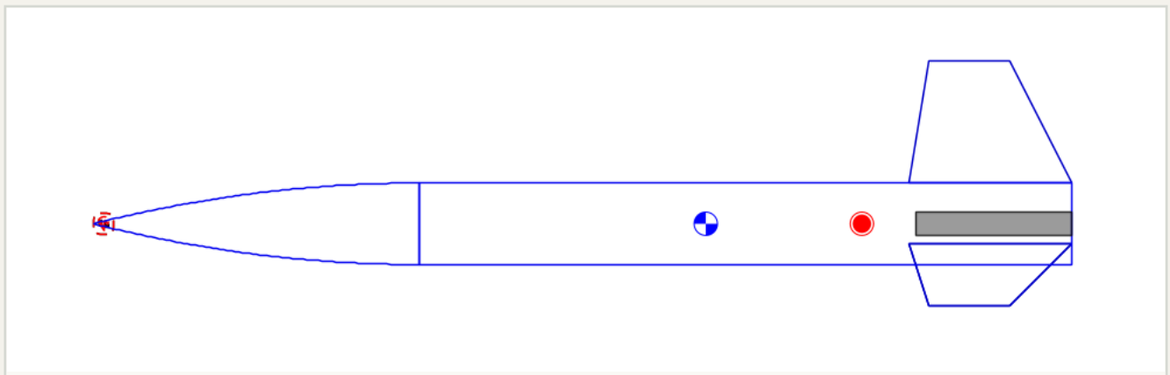


PERSONAL PROJECT / DECEMBER 2025

Butterfly valve assembly

A complete CATIA assembly exploring multi-part component fitment and kinematic movement.

CATIA / ASSEMBLY DESIGN / KINEMATICS



PERSONAL PROJECT / IN PROGRESS

L1 solid rocket design & build

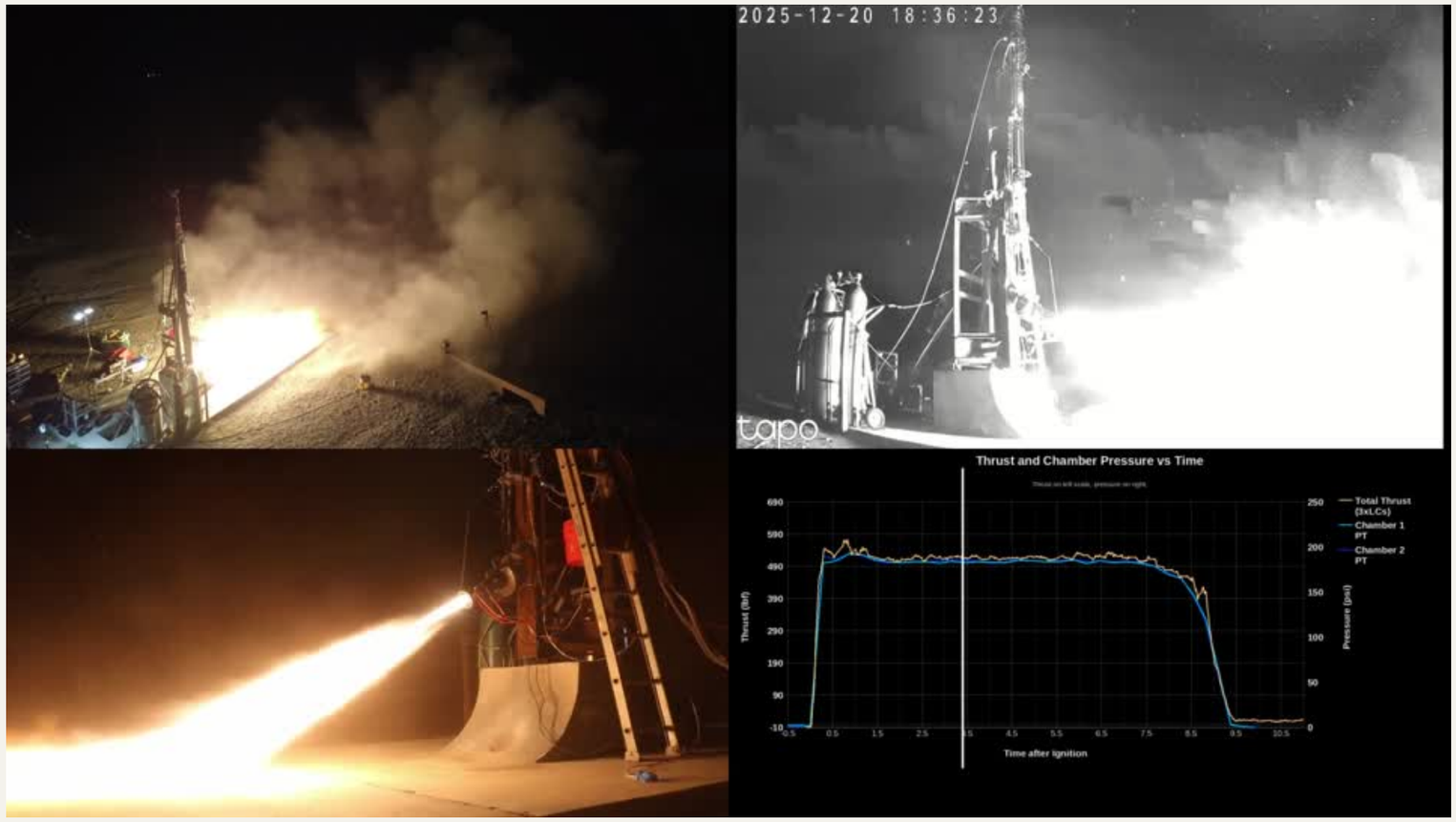
Rocket design for an H-class motor, with fiberglass fabrication and OpenRocket flight simulation.

OPENROCKET / COMPOSITES / FABRICATION

02 / TEST CAMPAIGN

Static fire testing & launch pad operations.

Beach Launch Team / California State University, Long Beach



STATIC FIRE / MULTIPLE CAMERA VIEWS + TEST DATA

A still from the test video combines camera views with thrust and chamber-pressure traces. Open the portfolio to play the full video.



BEACH LAUNCH TEAM / AT THE PAD

The team with the vehicle and ground infrastructure during field operations.

My contribution

Aerostructures pre-test checks, structural vehicle integration, and launch pad operations as part of the Beach Launch Team.

Pre-test preparation

Participated in structural vehicle integration and pre-test checks, contributing to a successful static fire test.

From hardware to operations

Hands-on team experience connecting structural work, vehicle integration, and launch pad operations.